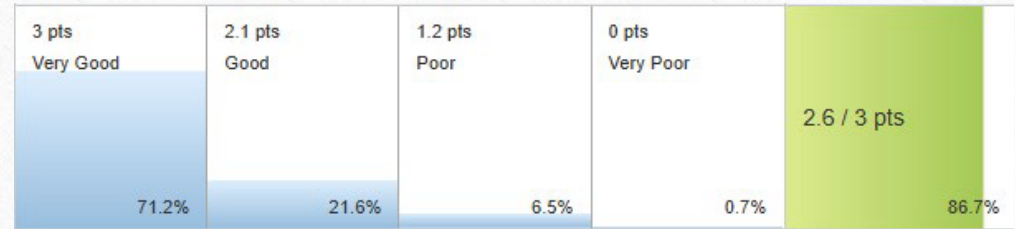


Iteration 2 ISAS Feedback

Infosys 722

1.1 -
Business/situati
on objectives
must be logical,
and in line with
data mining
goals and the
business success
criteria. Must be
conveyed in a
clear manner.

1.1



- Overall done well, with most students achieving full marks.
- Submissions demonstrated unique and interesting ideas.
- Some lacked explanation on how to assess achievement of business objectives or their alignment with data mining goals.

1.2 - Situation assessment must effectively describe the resources, requirements, assumptions, constraints, risks and contingencies of the project.

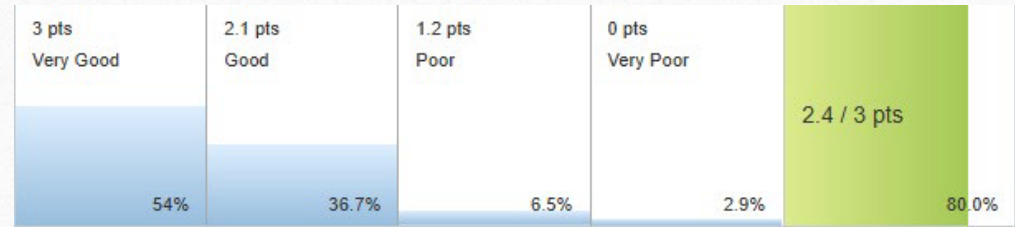
1.2



- Overall, the reports were strong, but many overlooked risks, and contingencies.

1.3 - Data mining goals must be achievable and closely aligned with the business objectives/success criteria. Must be conveyed in a clear manner.

1.3



- Future work should consider using **SMART goals** and defining more specific data mining model performance success criteria.

1.4 - The project plan must address how each phase of the project will be carried out for the current iteration. A day-to-day timeline must be proposed within the project plan.

1.4



- It was great to see that most reports included a day-to-day timeline.
- Stronger timelines provided more precise and insightful details, often supported by tables or calendar-style formats.
- Weaker timelines relied on vague phases and lacked clarity.
- To improve further, timelines should use actual dates (e.g., *8 July*) rather than generic labels like *Day 1*, *Day 2*, as projects typically operate with fixed deadlines.

2.1 - Collect initial data and describe where the data was collected from, how it was collected and any issues encountered during collection.

2.1



- Some students missed out describing ‘any issues encountered during collection’. We are simply looking for insight and sensitivity towards your data here. Does not have to be complex issues
- Remember to cite correctly, and describe how it was collected

2.2 – Data description must describe the format, quantity, fields and surface-level features of the data.

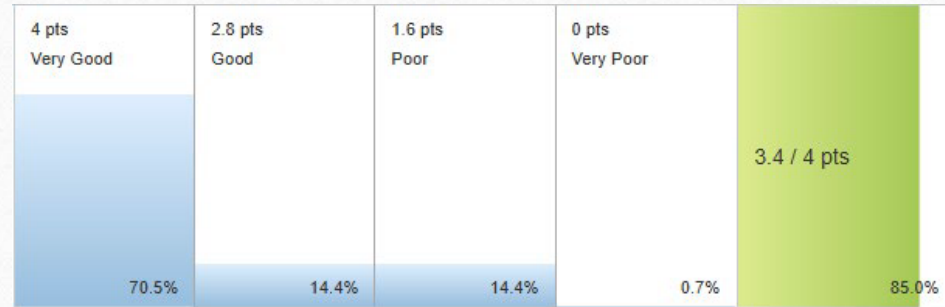
2.2



- Requires description of format, quantity and surface-level features
- Most students failed to describe the surface-level features of the data. Note that this involves describing the labels or headers of the field. Some labels are unclear therefore some further clarification should be made here.

2.3 - Data exploration must assist readers in understanding the data through the usage of strong visualisations (visualising the raw data). Must be communicated in a clear manner and explicitly linked to the rest of the steps.

2.3



- 'Usage of strong visualisations' pertains to the raw data.
- Think of this section as a preliminary investigation of your data, confirming its fitness for data mining
- What are the initial data patterns you can observe?
- Make sure to truly flesh out this logic from the screenshots rather than have the reader infer your logic

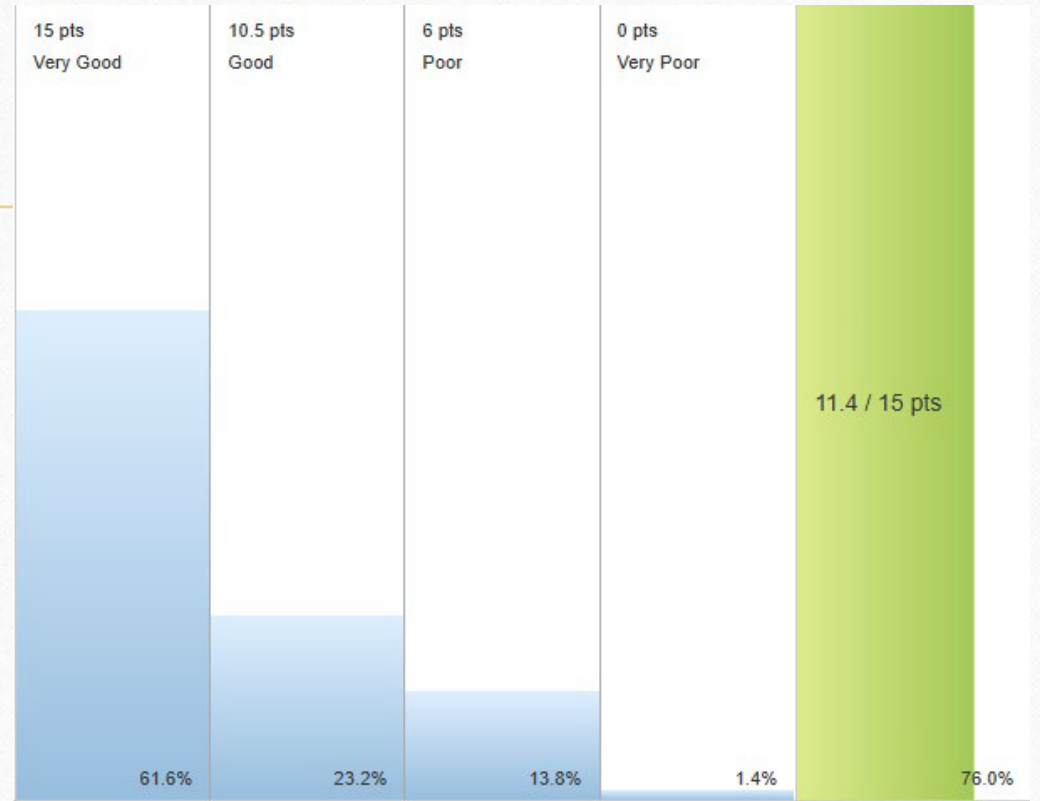
2.4 - Data quality must be verified by checking for errors, missing values, and data quality patterns explicitly.

2.4



- An extension of data exploration, but a closer look at data quality
- Note that data quality patterns can be observed in 2.3, but it is analysed deeper in 2.4
- Remember to consider errors, missing values and other data quality patterns
- Make some commentary on the overall cleanliness or quality of your data through this discussion

3. Data Preparation As a general guide: 3.1 - Data must be selected effectively. Goals, data quality, technical constraints, among other factors should be explicitly considered. 3.2 - To clean the data, issues must be made explicit, then explicitly resolved. 3.3 - Data must be appropriately constructed through the creation of new features/variables, and/or data repositories/tables. 3.4 - Integration must take place. This includes effectively merging data from various sources. 3.5 - Reformatting includes changing the formats of different data sources and trimming content, among other steps specific to the data.



3. Data
Preparation As a
general guide:
3.1 - Data must
be selected
effectively.
Goals, data
quality, technical
constraints,
among other
factors should
be explicitly
considered. 3.2 -

3.1

- Variable selection needs clearer justification: explanations for their use are required.
- Explanations should explicitly address how choices match data mining goals, account for data quality, and consider technical constraints in reports.

considered. 3.2 -
To clean the
data, issues
must be made
explicit, then
explicitly
resolved. 3.3 -

3.2

- Data issues ‘must be made explicit’ Refer and reidentify the issues that need to be resolved
- List them out with direct reference from section 2
- Show exactly how you dealt with them
- If the dataset is already completely clean, consider adding some “dirty” data (e.g., outliers or empty records) so the cleaning step can be demonstrated. Using completely clean data without showing any cleaning will result in losing marks for this section.

resolved. 3.3 -
Data must be
appropriately
constructed
through the
creation of new
features/variables,
and/or data
repositories/tables.
3.4 -

3.3

- ‘Must be appropriately constructed’
- Create a new variable, changing the number into a group, such as putting age into groups.

es. 3.4 -
Integration must
take place. This
includes
effectively
merging data
from various
sources. 3.5 -

3.4

- Integration must be demonstrated. Many students lose marks in this section, so if you do not have multiple sources, you should split your dataset in advance and recombine it here.
- Since this course focuses on the data mining process, it is important to experience these steps for the benefit of your own learning.
- An easy way to meet this requirement is to split your dataset into two parts and then integrate them again.

sources. 3.5 -
Reformatting
includes
changing the
formats of
different data
sources and
trimming
content, among
other steps
specific to the
data.

3.5

- Acknowledge the section even if it is not necessary
- Acknowledge this section even if minimal changes are required.
- Even small reformatting tasks count, such as converting a variable from *numeric* to *datetime*.
- Reformatting is usually straightforward, and most students handled it well. Just remember to explain your approach. If your data format is already excellent (which is rare), make that clear too.

4.1 - Data must be reduced through the selection of features relevant to the predictor (horizontal reduction) and/or vertical reduction. This could be achieved through the use of feature selection/logical processes.

4.1



- Feature selection - the most common reduction task
- Reiteration possibilities - trying out different variables for your algorithm

4.2 - The data must be projected through the use of statistical transformations (such as taking the log of a distribution).

4.2



- Provide some insight into potential biases that may still remain in your data, even after Section 3 (e.g., unresolved issues).
- This stage often involves applying log scaling to address skewness in large datasets.
- Log transformations can reduce the impact of extreme observations that persist after cleaning or imputation.

5.1

5 pts Very Good	3.5 pts Good	2 pts Poor	0 pts Very Poor	3 / 5 pts	
21.9%	49.6%	20.4%	8%		60.0%

5.1 - Match and discuss DM methods within the context of the DM objectives.

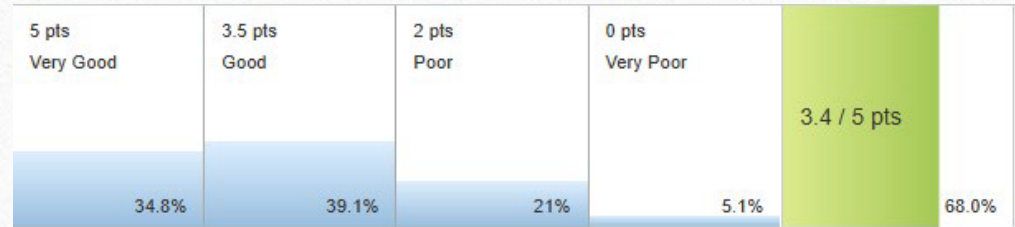
- 5.1 and 5.2 work hand-in-hand
- **‘Match and discuss’**

Please refer to the “building the model” guidelines linked below to ensure you cover the required discussion points for this section:

<https://www.dataquest.io/blog/sci-kit-learn-tutorial/>

5.2 - Select the appropriate DM method(s) in a logical manner. The selected DM method must be in line with the data mining goal/success criteria.

5.2



- ‘Appropriate’ DM method(s) are based on your context and your data mining goals.
- It would be useful to concurrently work on section 5 and section 1 (data mining goals and success criteria) so that you can provide an aligned, logical justification for your selection
- You can use the nature of your data as part of your justification.

6.1 Conduct exploratory analysis of DM algorithms within the context of the DM objectives. Then, discuss the analysis.

6.1



- This section mirrors the discussion in Section 5 but is applied at the **algorithm level**.
- You can focus on the conceptual differences between algorithms.
- Comparisons can also be made based on **accuracy levels**, though it is best to integrate this within your broader documentation and reasoning for clarity.

6.2 - Select algorithm(s) in a logical manner based on the exploratory analysis and discussion.

6.2



- Same as 5.2 but at the algorithm level
- Explicitly describe your choice of a suitable algorithm, such as linear regression or two-step clustering, etc. and justify your choice.

6.3 - Model(s) must be selected/built, and the appropriate algorithm/model parameter(s) must be selected.

6.3



- Most students successfully built or selected models, but many overlooked **parameterisation in Section 6.3**.
- Retuning parameters can be documented effectively by dividing into iteration phases (e.g., *Phase 1 vs Phase 2*) and showing which produced the most appropriate results.

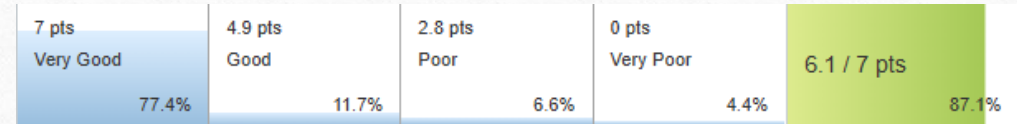
7.1 - Logical test designs must be created. Justify why a particular test design was used (for example, why was a 70/30 training/testing split used?).

7.1



- Some robust discussion for the split would be good here
- Usually, the students who scored higher in this section were able to cite studies arguing why the 70/30 split is a good rule of thumb/test design for most data mining methods.
- Synthesising cited documentation usually produced excellent justifications. Please suggest why we are using that kind of split. Is there any reference to back up the choice?

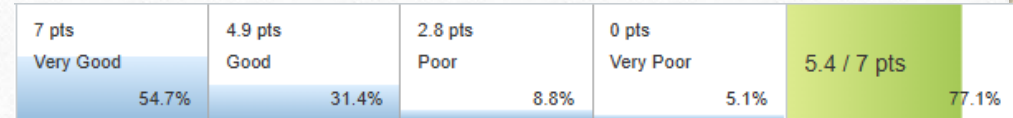
7.2



7.2 - Data mining must be conducted (the model must run).

- The most important section of them all.
- Your model must execute, **otherwise you will receive 0**

7.3



7.3 - Search for patterns and document the model's output.

- Remember the 'data exploration' section where you had to do some preliminary observation of the data?
- 7.3 is similar, however, within the context of the model's output. Can very much overlap with 8.1 and 8.2.
- This can be as simple as presenting a basic linear regression that demonstrates a positive linear trend.

8.1 - Study and discuss the mined patterns. Carry out an in-depth discussion about the data, results, models and patterns.

8.1



- Linked to 7.3
- A preliminary study/discussion of the mined patterns can help determine the validity or reliability of your data mining

8.2 - Visualise
the data, results,
models and
patterns in a
clear and
effective
manner.

8.2



- Ties in with the whole section 8
- So long as there are effective, legible and clear visualisations of the data, results, models and patterns, you should do well for this section.

8.3 - Interpret
the results,
models and
patterns
showing a clear
understanding
of the results.

8.3



- Your interpretation is based on the nature of your method/algorithm
- If you are unfamiliar with how to interpret your models/patterns, we encourage you to look at similar studies and interpret equivalently.
- Remember to directly refer to the values, features and elements of your models/patterns within your interpretation to make it effective.

8.4 - Assess and evaluate the results, models and patterns using the appropriate methods/processes.

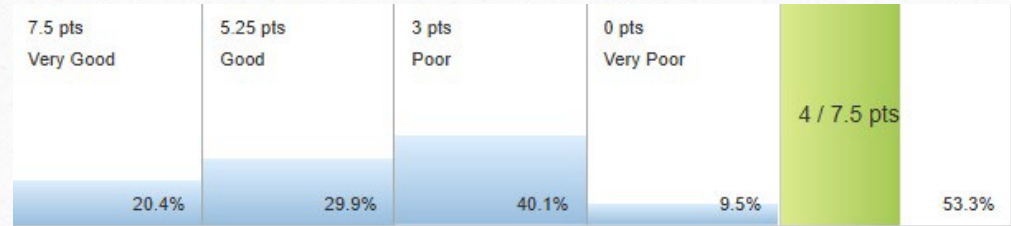
8.4



- **“Assess and evaluate” – what is the quality, value and importance of your data mining? To what degree were your goals attained?**
- It's often smart to evaluate against a standard that you set in your project. The best way to do this is to compare your findings to your initial data mining goals, hypotheses, and success criteria.

8.5 - Multiple iterations must take place. Repeat various steps to ensure your model is effective/robust. Document the steps/processes /reasoning.

8.5



- Students who did well in this section showed a careful and sensitive reiteration of steps to ensure that the models are **effective or robust**.
- The best way to do this is to show throughout your document multiple phases and reiterations within each section. For example, phase 1 vs phase 2.
- Multiple iterations must take place, play around with your data with different settings and see any changes. For example, if you removed outliers before, put them back and see the difference.



Thank you!!